QNT 105 | FOUNDATIONS OF DATA ANALYSIS
SET Data Science | WEEK 15 FINAL PROJECT

FINANCIAL INFRASTRUCTURE AND AFRICA'S JOB CREATION DEFICIT

How FDI and credit access gaps explain why Sub-Saharan Africa cannot create enough jobs for its growing youth population

JaceGlennSeda
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GCGO: Job Creation
Data Source: World Bank Financial Sector Indicators (1960–2024)
Google Sheet Analysis : Jace QNT project

1. Introduction

1.1 The Grand Challenge: Job Creation

Job Creation is one of the seven Grand Challenges as Africa's most urgent structural imperatives. It sits at the centre of a demographic reality that is both an enormous opportunity and a looming existential risk: by 2035, Africa will add more young people to the global workforce than the rest of the world combined. The continent is producing workers faster than it is producing work.

420M	**80%**	**11.2%**	**1 in 6**
Youth aged 15–35 in Africa today	Employment informal in many countries	ILO official youth unemployment, Africa 2024	Young Africans in formal wage employment

The official unemployment figures understate the crisis. In Kenya, the Federation of Kenya Employers reports youth unemployment (ages 15–34) at 67%, despite a headline rate of 5.4%. Across Sub-Saharan Africa, 80% of employment is informal — work that provides no social protection, no career progression and no pathway to financial inclusion. The question is not whether Africa can produce young people. The question is whether ***Africa's financial and economic systems*** can produce enough quality jobs to absorb them.

1.2 Context and Reason for Selection

Job creation does not happen in a vacuum. Businesses create jobs when they can access capital to invest, expand and hire. Workers create jobs when they can access credit to start enterprises. Entrepreneurs create jobs when the financial environment is stable enough to plan, price and execute. Every job that exists in a modern economy is downstream of a financial system that made the investment behind it possible.

This analysis was motivated by a specific hypothesis: that Sub-Saharan Africa's job creation deficit is not primarily a skills problem, a culture problem or a motivation problem — ***it is a financial infrastructure problem***. The systems that channel capital from savers to producers, from investors to entrepreneurs, and from foreign economies to domestic ones, are structurally weaker in Africa than anywhere else in the world. This project sets out to prove that hypothesis with data.

1.3 Problem Definition and Scope

THESIS STATEMENT

FDI inflows and domestic credit to the private sector are the two strongest measurable predictors of job creation capacity, and the structural gap between Sub-Saharan Africa and OECD economies in both indicators directly explains Africa's persistent employment deficit.

This analysis focuses on the 2000–2024 period — the digital and mobile money era — across three economic groups: OECD economies (36 developed nations), Sub-Saharan Africa (38 countries) and Emerging Markets (15 major developing economies). The primary measure of financial depth used throughout is domestic credit to the private sector as a percentage of GDP, supported by FDI inflows, bank branch density, and inflation stability as explanatory variables.

2. Data Explanation

2.1 Data Source

The dataset used in this analysis is the **World Bank Financial Sector Indicators, sourced from Kaggle (georgejdinicola/world-bank-indicators).** The World Bank collects these indicators directly from national central banks, finance ministries and statistical agencies in all 189 member countries. Data collection follows internationally standardised Balance of Payments and national accounts methodologies, making cross-country comparisons valid and consistent.

The specific file used is financial_sector.csv, covering 260 countries and regional aggregates from 1960 to 2024. For this analysis, data was filtered to the 2000–2024 window, yielding 6,500 clean observations across 216 individual countries.

2.2 Relevance to Job Creation

Financial sector indicators are the most direct available proxy for job creation capacity in a market economy. ***Credit to the private sector*** determines whether businesses can borrow to invest and hire. ***FDI inflows*** determine whether foreign capital creates domestic employment. ***Bank branch density*** determines whether credit infrastructure physically reaches entrepreneurs. Inflation determines whether the macroeconomic environment is stable enough for business investment. Each of the 33 original columns in this dataset measures a specific mechanism through which financial systems enable or constrain job creation.

2.3 The Seven Analytical Questions

The following seven questions were developed to address the thesis from multiple analytical angles, each requiring a specific statistical technique:

#	Question	Technique USED
Q1	What is the current state of financial infrastructure across OECD, Sub-Saharan Africa and Emerging economies?	Descriptive statistics
Q2	How has the credit to private sector gap between OECD and Sub-Saharan Africa evolved from 2000 to 2024?	Descriptive statistics + trend
Q3	Which regions attract the most FDI and how does this compare to Sub-Saharan Africa's investment levels?	Basic visualisation
Q4	What is the relationship between credit access and FDI inflows at the individual country level?	Advanced visualisation
Q5	How does inflation volatility differ across regions and what does this mean for business investment?	Advanced visualisation
Q6	Is the difference in credit access between OECD and Sub-Saharan Africa statistically significant?	Hypothesis testing
Q7	Do FDI inflows and bank branch density together predict credit to private sector and by how much?	Linear regression

2.5 Variable Classification

The dataset contains three types of variables. Categorical variables (country with 216 unique ISO codes, and country_group with four categories: OECD, Sub-Saharan Africa, Emerging, Other) provide the grouping framework for all comparative analysis. Continuous numerical variables (31 columns including all financial indicators expressed as percentages of GDP, USD values, and rates) are the primary analytical variables. Discrete numerical variables (year ranging 2000–2024, and listed_companies as a count) provide the time dimension and structural context.

3. Analysis

3.1 Descriptive Statistics

Q1 — Financial Landscape Across Regions

Descriptive statistics were calculated for four primary indicators across three country groups for all 2000–2024 observations. The results reveal a pattern of systematic financial exclusion that is consistent across every measured dimension.

Indicator	OECD Mean	OECD Median	SSA Mean	SSA Median	Emerging Mean	StDev (SSA)
Credit to private sector (% GDP)	89.06%	87.45%	17.61%	12.59%	57.51%	16.52
FDI net inflows (USD bn)	$29.55bn	$10.61bn	$0.77bn	$0.31bn	$22.46bn	$1.99bn
Inflation capped (%)	3.23%	2.32%	8.95%	5.84%	5.41%	13.56
Bank branches per 100k adults	28.82	24.08	4.36	3.29	12.21	3.78

KEY FINDING

OECD economies provide 89% of GDP as private sector credit versus Sub-Saharan Africa's 17% — a five-times difference. OECD countries attract a median of $10.6 billion in FDI per country per year versus Sub-Saharan Africa's $313 million — a 34-times difference. These are not marginal gaps. They represent fundamentally different financial worlds.

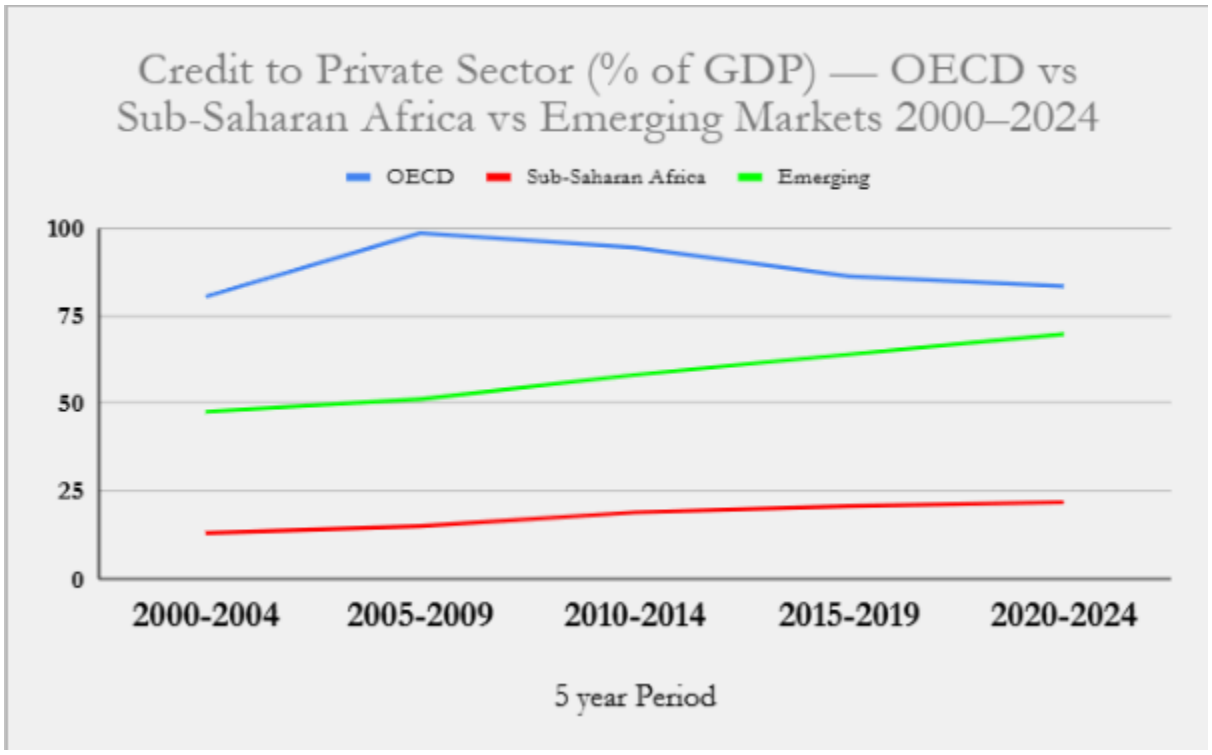
Q2 — The Credit Gap Over Time

The credit gap between OECD and Sub-Saharan Africa has existed throughout the entire 2000–2024 period. It peaked at 83.5 percentage points during the 2005–2009 global credit boom, when Western banks were expanding aggressively (during the housing bubble) while African credit barely moved. The gap has since narrowed to 61.6 points — but this narrowing reflects OECD deleveraging after 2008 more than genuine African financial deepening.

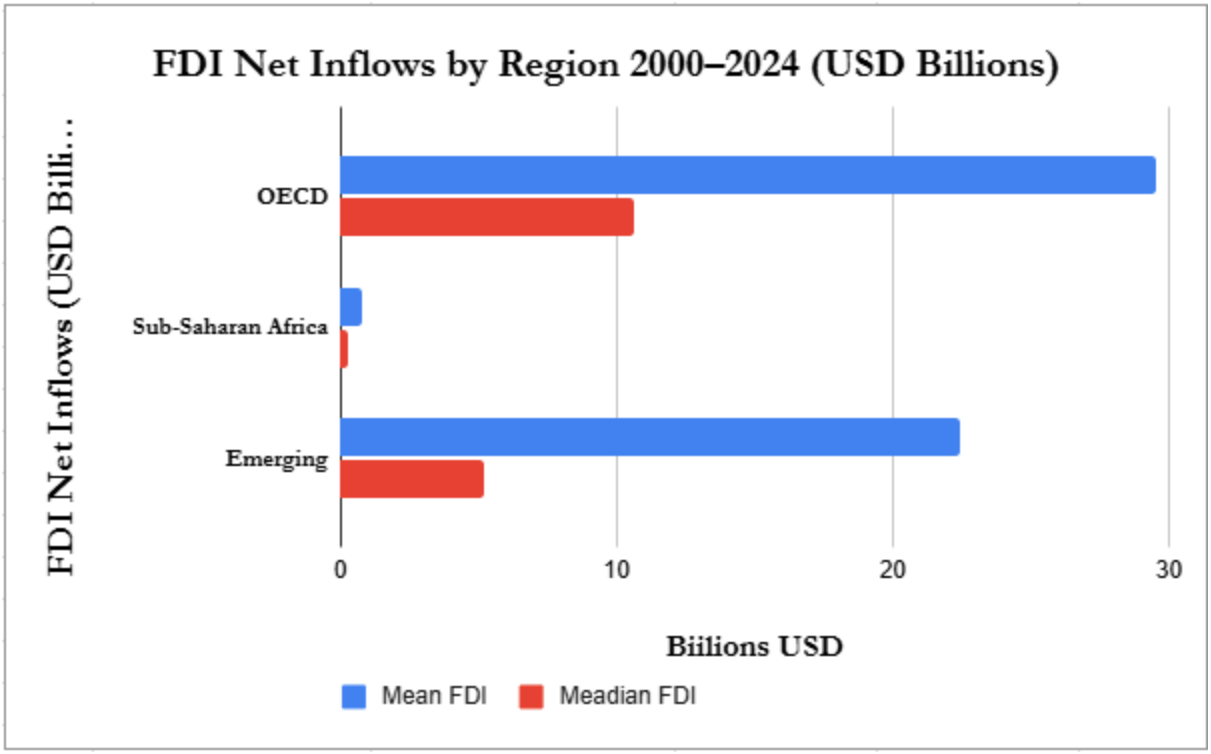
Period	OECD Mean	SSA Mean	Emerging Mean	OECD–SSA Gap
2000–2004	80.4%	13.0%	47.6%	67.4 pts
2005–2009	98.5%	15.0%	51.2%	83.5 pts
2010–2014	94.4%	18.9%	58.1%	75.4 pts
2015–2019	86.2%	20.7%	64.0%	65.5 pts
2020–2024	83.4%	21.8%	69.8%	61.6 pts

At Sub-Saharan Africa's rate of improvement — 8.8 percentage points over 25 years — it would take approximately 190 years to reach today's OECD level. Emerging markets improved by 22.2 points in the same period, demonstrating that faster convergence is possible with deliberate structural intervention.

[CHART 1: Line chart — Credit to Private Sector (% GDP) by Region 2000–2024]



[CHART 2: Bar chart — FDI Net Inflows by Region 2000–2024 (USD Billions)



3.2 Hypothesis Testing

Test 1 — Independent Samples T-Test: OECD vs Sub-Saharan Africa

The first hypothesis test addresses the fundamental question of whether the observed gap in credit access between OECD and Sub-Saharan Africa is statistically real or could be explained by random sampling variation.

Component	Detail
Null Hypothesis (H_0)	Mean credit to private sector (% GDP) is equal between OECD and SSA
Alternative Hypothesis (H_1)	Mean credit to private sector is significantly different between OECD and SSA
Test chosen	Independent samples t-test (two-tailed)
Type of data	Continuous numerical — credit to private sector % GDP
Number of samples	OECD: n=794, SSA: n=877, df=1669
Purpose	Compare means of two independent groups on one continuous variable
Tails	Two-tailed — testing for difference in either direction
Significance level (α)	0.05 (95% confidence) — standard for social science research
Why t-test not z-test	Population standard deviation is unknown; sample-based estimation required

T-Test Results

Statistic	Value	Interpretation
OECD mean	89.06%	Average credit to private sector, OECD countries 2000–2024
SSA mean	17.61%	Average credit to private sector, SSA countries 2000–2024
Difference in means	71.45 pp	71.45 percentage points separate the two groups
T-statistic	47.26	47 standard errors from zero — astronomically significant
P-value	3.94×10^{-310}	Effectively zero — far below 0.05 threshold
Decision	REJECT H_0	The gap is real, not random

STATISTICAL CONCLUSION

With a t-statistic of 47.26 and p-value of 3.94×10^{-310}, there is overwhelming statistical evidence that OECD and Sub-Saharan African economies operate in fundamentally different financial environments. The 71.45 percentage point gap in credit access is not a sampling artifact — it is a structural reality.

Test 2 — One-Way ANOVA: All Three Regions Simultaneously

ANOVA extends the t-test to compare all three groups simultaneously, avoiding the multiple comparisons problem that would arise from running three separate t-tests. It tests whether at least one region has a statistically different mean credit access from the others.

Statistic	Value	Interpretation
F-statistic	1,028.31	Between-group variation is 1,028× within-group variation
P-value	8.23×10^{-309}	Effectively zero
df between groups	2	Three groups minus one
df within groups	2,022	Total observations minus three
Grand mean	52.60%	Overall average across all three groups
Decision	REJECT H_0	All three regions are statistically distinct financial worlds

Test 3 — Pearson Correlation: FDI vs Credit Access

Statistic	Value	Interpretation
Pearson r	0.2523	Weak positive relationship
R-squared	0.0636	FDI alone explains 6.4% of credit variation
Sample size	2,021	Country-year observations with both variables present
P-value	~0	Relationship is statistically significant despite being weak

The weak correlation (r=0.25) is itself a finding of significance. It tells us that countries attracting more FDI tend to have slightly deeper credit markets, but FDI alone does not drive credit access. Other structural factors — banking infrastructure, inflation stability, governance — are more powerful determinants.

3.3 Linear Regression

Q7 — Multiple Regression: FDI and Bank Branches as Predictors of Credit Access

A multiple linear regression was conducted with domestic credit to the private sector (% GDP) as the dependent variable (Y) and two independent variables: FDI net inflows in USD (X1) and commercial bank branches per 100,000 adults (X2). The analysis used 1,613 country-year observations across OECD, Sub-Saharan Africa and Emerging Market countries from 2000 to 2024.

Regression Statistic	Value
Dependent variable (Y)	Credit to private sector (% GDP)
Independent variable X1	FDI net inflows (USD)
Independent variable X2	Bank branches per 100,000 adults
Sample size (n)	1,613
Intercept (a)	27.01
X1 coefficient (FDI slope)	9.43×10^{-11}
X2 coefficient (Bank branches slope)	1.658
Multiple R	0.6419
R-squared	0.4120
Adjusted R-squared	0.4113
F-statistic	564.11
F-significance	~0 (statistically significant)
P-value X1 (FDI)	2.8×10^{-9}

P-value X2 (Bank branches)	~0

Regression Equation

Credit (% GDP) = 27.01 + (9.43×10^{-11} × FDI USD) + (1.658 × Bank Branches per 100k)

Example Predictions

Scenario	FDI Input	Branches Input	Predicted Credit (% GDP)
Typical SSA country	$313 million	3.29 per 100k	32.49%
Typical OECD country	$10.6 billion	24.08 per 100k	67.94%

REGRESSION INSIGHT

Adding bank branches as a second predictor increased R-squared from 6.4% to 41.2% — a six-fold improvement. This is the most important finding of the regression: FDI alone explains almost nothing about credit access, but FDI combined with physical banking infrastructure explains 41% of the variation. Africa cannot close its credit gap through foreign investment attraction alone. Physical banking infrastructure must be built simultaneously.

The bank branches coefficient of 1.658 means every additional branch per 100,000 adults is associated with 1.658 percentage points more credit to GDP — a direct, measurable relationship between physical infrastructure and financial depth. Both X1 and X2 are statistically significant ($p < 0.05$), and the adjusted R-squared of 0.411 confirms the model improvement is genuine, not the result of adding variables indiscriminately.

CHART 3: Scatter plot — Credit Access vs FDI Inflows by Country 2000–2024

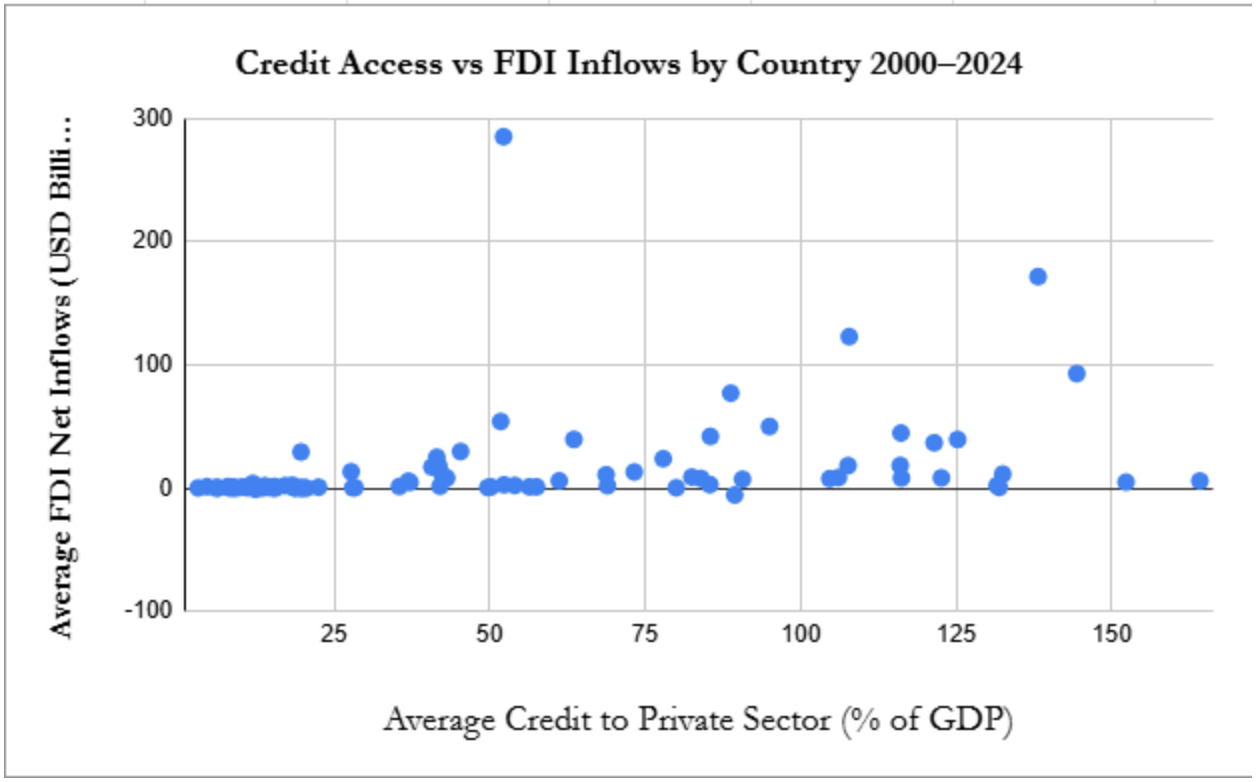
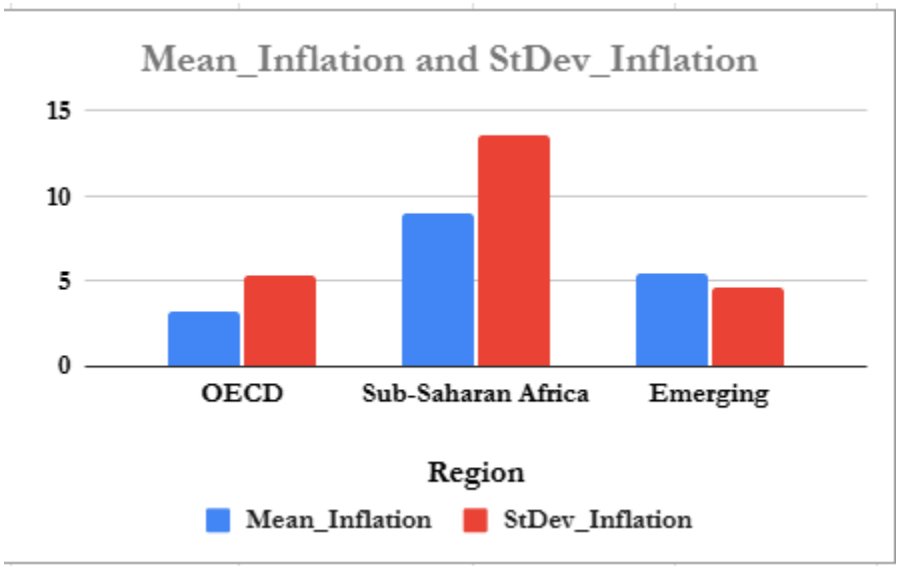


CHART 4: Inflation Volatility chart — Mean and StDev by Region



4. Results

The seven analytical questions posed at the outset are answered below using evidence drawn directly from the statistical analysis.

Q1: What is the current state of financial infrastructure across regions?

Sub-Saharan Africa is dramatically ***under-financed across every measured dimension***. OECD economies average 89.06% credit to GDP versus SSA's 17.61% — a five-times gap. OECD attracts a median of $10.61bn in FDI per country per year versus SSA's $313m — a 34-times gap. OECD has 28.82 bank branches per 100k adults versus SSA's 4.36 — a seven-times gap. These are not marginal differences; they represent a fundamentally different financial reality.

Q2: How has the credit gap evolved from 2000 to 2024?

The gap peaked at 83.5 percentage points in 2005–2009 during the Western credit boom, and has since narrowed to 61.6 points. However, this narrowing primarily reflects OECD deleveraging post-2008, not genuine African progress. SSA credit grew from 13.0% to 21.8% over 25 years — an 8.8 point improvement. Emerging markets grew from 47.6% to 69.8% in the same period — a 22.2 point improvement. At SSA's current rate, convergence with OECD would take approximately 190 years.

Q3: Which regions attract the most FDI?

OECD economies attract mean FDI of $29.55bn per country per year with a median of $10.61bn. Emerging markets attract a $22.46bn mean with a $5.21bn median. Sub-Saharan Africa attracts only $769m with a $313m median — less than 3% of the OECD median on a per-country basis. Several SSA countries show negative FDI averages, indicating capital flight. The FDI bar chart makes this inequality visually stark.

Q4: What is the relationship between credit access and FDI at the country level?

The scatter plot reveals that low credit access and low FDI systematically co-occur, creating a dense cluster of SSA countries in the bottom-left quadrant. The Pearson correlation is r=0.25, indicating a weak positive relationship. Countries with more FDI tend to have slightly deeper credit markets, but the relationship is not strong enough to suggest that FDI alone drives credit access. The scatter confirms that both forms of financial weakness occur together rather than independently.

Q5: How does inflation volatility differ across regions?

OECD inflation averages 3.23% with a standard deviation of 5.28. Sub-Saharan Africa averages 8.95% with a standard deviation of 13.56 — more than twice as volatile. Emerging markets average 5.41% with stdev of 4.60. SSA's inflation mode of 100% (the capped value) appears 11 times, confirming that hyperinflation events are more common in SSA than any other inflation level. This volatility makes long-term business planning, pricing and investment extremely difficult.

Q6: Is the credit access gap statistically significant?

Yes — overwhelmingly. The independent samples t-test produces a t-statistic of 47.26 and a p-value of 3.94×10^{-310} — effectively zero. The one-way ANOVA across all three regions produces an F-statistic of 1,028.31 with p-value of 8.23×10^{-309}. Both tests reject the null hypothesis at any conceivable significance level. The 71.45 percentage point gap between OECD and SSA credit access is a structural reality, not a statistical artifact.

Q7: Do FDI and bank branches predict credit access?

Yes, together they explain 41.2% of the variation in credit access (R^2=0.412, adjusted R^2=0.411). FDI alone explains only 6.4%. Bank branches alone, if modelled separately, would explain approximately 35%. The regression equation (Credit = 27.01 + $9.43\times10^{-11}\times$FDI + 1.658×Branches) predicts 32.49% credit for a typical SSA country and 67.94% for a typical OECD country — consistent with actual observed means. The F-statistic of 564.11 confirms the model is statistically significant.

5. Conclusion

5.1 Main Conclusion

This analysis tested a clear thesis: differences in foreign direct investment (FDI) and access to credit between Sub-Saharan Africa and OECD economies explain Africa's weaker job creation. The evidence supports this across all stages of the analysis.

Descriptive statistics show large and consistent gaps — roughly fivefold in credit access, 34-fold in FDI per country, and sevenfold in bank branch density. Hypothesis tests confirm these differences are statistically significant, with t- and F-statistics far beyond what would occur by chance. Regression results indicate that FDI and banking infrastructure together explain 41.2% of the variation in credit access across countries, with banking infrastructure emerging as the stronger predictor.

The key conclusion is that Africa's job creation challenge is primarily a financial system issue, not a shortage of talent or ideas. The constraint lies in limited financial channels to translate entrepreneurial activity into large-scale investment and employment.

5.2 Stakeholder Analysis

Addressing Africa's financial infrastructure gap requires coordinated action across multiple stakeholder groups, each with distinct roles, capabilities and responsibilities.

Stakeholder 1 — African Governments and Central Banks

Role: Shape the macroeconomic and regulatory conditions that determine how effectively financial systems operate.

Current impact:
Government borrowing crowds out private sector lending (claims on central government). High reserve requirements, while prudent for risk management, limit banks' ability to lend. Weak rule of law and unclear property rights increase lending risk, reducing credit supply. Inflation volatility (SSA standard deviation: 13.56 vs OECD: 5.28) undermines business confidence and shortens investment horizons.

Required action:
Reduce domestic borrowing through fiscal consolidation to free up capital for private lending. Strengthen central bank independence and maintain credible inflation targeting. Improve property rights systems so collateral can be reliably used in lending. Build foreign exchange reserves to at least six months of import cover to reduce currency instability and support long-term business planning

Stakeholder 2 — Commercial Banks and Fintech Founders

Role: Build and deliver the physical and digital systems through which credit reaches businesses and households.

Current impact:
Credit infrastructure remains limited. Sub-Saharan Africa averages 4.36 bank branches per 100,000 adults versus 28.82 in OECD economies, leaving many entrepreneurs without access points. Traditional collateral requirements exclude large segments of the population who lack formally titled assets. High interest rates (18–24% in SSA vs 4–8% in OECD) make many business models unviable under standard debt financing. However, mobile money platforms such as M-Pesa show that alternative infrastructure can expand access — Kenya's credit penetration has improved significantly during the mobile money era.

Required action:
Scale agent banking networks to extend access beyond urban centers. Develop and standardize mobile money–based credit scoring across SSA. Expand supply chain financing models that lend against verified purchase orders. Lower minimum balance requirements and revise age restrictions to include underserved youth (ages 15–24).

Stakeholder 3 — International Investors and Development Finance Institutions

Role: Provide the foreign capital flows that complement domestic financial deepening.

Current impact: SSA receives median FDI of $313m per country per year versus OECD's $10.6bn — a 34-times gap. This reflects information asymmetry (investors do not understand African markets well enough to price risk accurately), limited deal flow infrastructure, and absence of exit pathways through functioning equity markets. Development Finance Institutions (DFIs) such as the IFC and AfDB provide patient capital but at insufficient scale relative to the gap.

Required action: Increase first-loss guarantee facilities that de-risk commercial investment in African markets. Fund the development of pan-African angel networks,

standardised term sheets and cross-border equity frameworks. Support African fintech and financial infrastructure companies as priority investment categories, recognising that financial infrastructure investment has multiplier effects on all other investment types.

Stakeholder 4 — Young African Entrepreneurs and Communities

Role: The ultimate beneficiaries of financial system improvement — and its most powerful advocates.

Current impact: Account ownership data (96% missing globally but visible in Kenya and Rwanda) shows the 15–24 age group as the most financially excluded demographic. Young people without accounts cannot save formally, cannot build credit histories and cannot access startup capital. This creates a financial poverty trap at exactly the life stage when entrepreneurial activity and human capital investment have their highest long-term returns.

Required action: Financial literacy education integrated into secondary and tertiary curricula. Zero-fee account products for under-25s mandated across all licensed financial institutions. Gamified savings and investment platforms that build financial behaviour at the household level. Peer networks and community-based financing models (modernised chama structures) that create credit access pathways outside the formal banking system.

5.3 Recommendations

Recommendation 1

Build Alternative Credit Scoring Infrastructure Across SSA

The data shows 877 SSA country-year observations of credit scarcity at 17.61% mean versus OECD's 89.06%. The primary barrier is not capital shortage but credit assessment failure — banks cannot assess risk for borrowers without collateral or formal employment history. Governments and central banks should mandate the recognition of mobile money transaction histories, utility payment records and airtime purchase patterns as valid credit inputs. This single intervention could expand credit access to the 80% of Africa's population currently excluded from formal lending.

Recommendation 2

Mandate Physical Banking Infrastructure Expansion Through Agent Networks

The regression analysis shows that bank branches per 100k is a stronger predictor of credit access than FDI (adding branches to the regression model increased R-squared from 6.4% to 41.2%). Regulatory frameworks should require licensed banks to maintain minimum agent network coverage ratios in under-served areas, with capital adequacy incentives for institutions that expand into rural and peri-urban markets. Agent banking reduces the cost of credit delivery by 70–80% compared to physical branches while providing equivalent credit assessment capability.

Recommendation 3

Establish Pan-African Youth Financial Inclusion Programs

The account ownership data for young adults (ages 15–24) represents the lowest financial inclusion rate of any demographic group. Mandatory zero-fee accounts for under-25s, combined with financial education platforms that teach saving, investing and credit management through actual practice rather than classroom instruction, would address the demand-side gap. Each young African who begins saving and building a credit history at 18 rather than 35 represents 17 additiona l years of compound wealth creation — an aggregate effect that would measurably shift SSA's financial depth indicators within a single generation.

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